

Lesson 6 — Build the Model & Count Loops

Matches LEGO Teacher Portal lesson: "Drop from the Top".

Objective (I can...): I can build the loops model and use a count-controlled loop to replace repeated blocks. **CS concept:** Iteration; count-controlled loops; a loop is abstraction (one "repeat N" stands in for N copies) · **Builds on:** Lesson 5 (sequences we now repeat efficiently).

AI Connection (2 min): Training an AI model is iteration at scale — the same update step runs over data thousands of times. *First-principles question:* "Why would repeating a tiny improvement many times produce a big result?"

Materials: LEGO CS & AI kit + device per pair, notebooks. **This is the unit's build lesson** — build the Loops model now and keep it assembled through Lesson 10.

60-minute flow - Warm-up (8): "Compress it" — coach writes out a long repeated instruction list; pairs rewrite it as the shortest possible instruction. Reveal: "repeat N times" is a count-controlled loop, and it's a form of abstraction. - **Concept + demo (10):** Show four copied drive blocks, then replace them with one block inside a repeat-4 loop; run both and confirm identical behavior. Point out how much less code can go wrong. - **Build/Code (30):** Pairs **build the Loops model** (the unit's one build), then program a count loop (drive → pause, repeated N times). They record in the notebook what one copied-out equivalent would look like. *Swap at 15 min.* - **Challenge (7):** Change the count and predict the total run time before testing; log prediction vs. actual. - **Share + pack-up (5):** Pairs show the loop, state the count, and explain why the loop version is easier to maintain. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator counts iterations aloud and logs the prediction. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a palette card highlighting loop blocks. *Extension / fast-finisher* — put two actions inside the loop body. *Reteach* — count iterations out loud together on the first run.

Exit check: Pair replaces repeated blocks with a working count loop and can explain the abstraction. **At-home (optional):** Find a repetitive task and write it as "repeat ___ times { ___ }." ~10 min.
